

Dichiarazioni: $Dec : D \rightarrow nil \mid const\ x : \tau = e \mid D_1; D_2 \mid D \text{ in } D \mid p \in Env$

Sintassi:

Semantica Statica $\Delta \in TE_{Env}$ $\Delta \vdash d_0 : \Delta_0$
 Assorio $\Delta_0 \in TE_{Env} \wedge d_0 \in Dec$

$\vdash nil : \emptyset$
 $\vdash p : v$ (v è l'insieme degli id a cui d_0 significa p)

$\Delta \vdash e : \tau$ $\Delta \vdash const\ x : \tau = e : [x \mapsto \tau]$
 Associa un tipo ad ogni espressione che non contiene errori di tipo nell'uso degli operatori

$\frac{\Delta \vdash d_1 : \Delta_1 \quad \Delta[\Delta_1] \vdash d_2 : \Delta_2}{\Delta \vdash d_1; d_2 : \Delta_1[\Delta_2]}$ $\frac{\Delta \vdash d_1 : \Delta_1 \quad \Delta[\Delta_1] \vdash d_2 : \Delta_2}{\Delta \vdash d_1 \text{ in } d_2 : \Delta_2}$

Esempio

$\Delta = \emptyset$
 (non abbiamo un ambiente contestuale)

$\begin{cases} const\ x : bool = true; & d_1 \\ const\ y : int = 5; & d_2 \\ const\ z : int = 6 * x; & d_3 \end{cases}$ $\frac{d_1; d_2; d_3}{d}$

d_1, d

per costruire l'ambiente statico di $d_1; d$
 dobbiamo costruire l'ambiente di d_1

$\vdash d_1; ? \rightarrow \dots$
 $\vdash d_1; d :$

① Elaboriamo d_1 statisticamente

$\frac{\vdash true : bool}{\vdash const\ x : bool = true : [x \mapsto bool]}$
 $\Rightarrow \frac{\vdash d_1 : [x \mapsto bool] \quad \emptyset[x \mapsto bool] \vdash d : ?}{\vdash d_1; d : ?}$

② Elaborazione d statisticamente

$\frac{[x \mapsto bool] \vdash d_2 : ?}{[x \mapsto bool] \vdash d_2; d_3 : ?}$

②a Elaborazione d_2 statisticamente

$\frac{[x \mapsto bool] \vdash 5 : int}{[x \mapsto bool] \vdash const\ x : int = 5 : [x \mapsto int]}$
 $\frac{[x \mapsto bool] \vdash d_2 : [x \mapsto int] \quad [x \mapsto bool][x \mapsto int] \vdash d_3 : ?}{[x \mapsto bool] \vdash d_2; d_3 : ?}$

②b

$\frac{[x \mapsto int] \vdash 6 * x : ?}{[x \mapsto int] \vdash const\ y : int = 6 * x : [y \mapsto int]}$

②c Valutiamo $6 * x$ statisticamente

$\frac{[x \mapsto int] \vdash 6 * x : int \quad [x \mapsto int] \vdash x : int}{[x \mapsto int] \vdash 6 * x : int}$

$\frac{[x \mapsto int] \vdash 6 * x : int}{[x \mapsto int] \vdash const\ y : int = 6 * x : [y \mapsto int]}$

$\frac{[x \mapsto bool] \vdash d_2 : [x \mapsto int] \quad [x \mapsto int] \vdash d_3 : [y \mapsto int]}{[x \mapsto bool] \vdash d_2; d_3 : [x \mapsto int][y \mapsto int]}$

$= [x \mapsto int, y \mapsto int]$ chiude il passo 2a e quindi il passo 2

Di conseguenza il passo ①

$\frac{\vdash d_1 : [x \mapsto bool] \quad [x \mapsto bool] \vdash d : [x \mapsto int, y \mapsto int]}{\vdash d_1; d : [x \mapsto bool][x \mapsto int, y \mapsto int]}$

Semantica dinamica $p \in Env$ $p \vdash d \rightarrow d'$
 $d, d' \in Dec$

$\vdash nil \rightarrow \emptyset$ $p \in Conf. terminali$

$\frac{p \vdash e \rightarrow v}{p \vdash const\ x : \tau = e \rightarrow [x \mapsto v]}$

$\frac{p \vdash d_1 \rightarrow d_1'}{p \vdash d_1; d_2 \rightarrow d_1'; d_2}$ $\frac{p \vdash d_2 \rightarrow d_2'}{p \vdash d_1; d_2 \rightarrow d_1; d_2'}$
 $p \vdash p_1; p_2 \rightarrow p, [p_2]$

$$\frac{\rho \vdash d_1 \rightarrow d_2}{\rho \vdash d_1 \text{ in } d_2 \rightarrow d_1' \text{ in } d_2} \quad \frac{\rho[p_1] \vdash d_2 \rightarrow d_2'}{\rho \vdash p_1 \text{ in } d_2 \rightarrow d_1' \text{ in } d_2'}$$

$$\rho \vdash p_1 \text{ in } p_2 \rightarrow \rho p_2$$

Elaborare (dinamicamente) una dichiarazione

Elab : Dec $\rightarrow$ Env

$\forall d \in \text{Dec} \quad \text{Elab}(d) = p \in \text{Env} \quad \text{sse}$

$\vdash d \vdash^* p$

Elab(d_1, d_2) = ambiente generato dall'applicazione delle regole della semantica dinamica

Equivalenza : $\equiv \in \text{Dec} \times \text{Dec}$

$d = d_2 \quad \text{sse} \quad \text{Elab}(d_1) = \text{Elab}(d_2)$

Esempio

$$\overbrace{\text{const } x : \text{int} = 2 \text{ in}}^{d_1} \underbrace{(\text{const } y : \text{int} = x + 1; \text{const } z : \text{int} = y + x)}_{d_2} \quad \overbrace{\text{const } z : \text{int} = y + x}^{d_3}$$

$d = d_2; d_3$

Primo passo

$d_1 \text{ in } d$

$$\frac{\vdash d_1 \rightarrow^* ?^{(1)}}{\vdash d_1 \text{ in } d \rightarrow ? \text{ in } d}$$

① $\vdash \text{const } x : \text{int} = 2 \rightarrow [x \mapsto 2]$ completa il passo 1

$$\frac{\vdash d_1 \rightarrow [x \mapsto 2]}{\vdash d_1 \text{ in } d \rightarrow [x \mapsto 2] \text{ in } d} \quad \text{Primo passo di elaborazione}$$

Secondo passo

$$\frac{[x \mapsto 2] \vdash d \rightarrow^* ?}{\vdash [x \mapsto 2] \text{ in } d \rightarrow [x \mapsto 2] \text{ in } ?}$$

Elaboriamo d : primo passo

$$\frac{[x \mapsto 2] \vdash d_2 \rightarrow}{[x \mapsto 2] \vdash d_2; d_3 \rightarrow}$$

Elaboriamo d_2

$$\frac{[x \mapsto 2] \vdash x + 1 \rightarrow^{(2)}}{[x \mapsto 2] \vdash \text{const } y : \text{int} = x + 1 \rightarrow}$$

② Valutazione di $x + 1$

$$\frac{[x \mapsto 2] \vdash x \mapsto 2}{[x \mapsto 2] \vdash x + 1 \rightarrow} \quad \text{perché } \rho(x) = 2$$

$$\bullet [x \mapsto 2] \vdash 2 + 1 \rightarrow 3 \quad \text{perché } 2 + 1 = 3$$

$\Rightarrow [x \mapsto 2] \vdash x + 1 \rightarrow 3$ completa la valutazione ②

$$\frac{[x \mapsto 2] \vdash \text{const } y : \text{int} = x + 1 \rightarrow [y \mapsto 3]}{[x \mapsto 2] \vdash \text{const } y : \text{int} = x + 1 \rightarrow [y \mapsto 3]} \quad \text{completa ③}$$

$$\frac{[x \mapsto 2] \vdash d_2 \rightarrow [y \mapsto 3]}{[x \mapsto 2] \vdash d_2; d_3 \rightarrow [y \mapsto 3]; d_3 \rightarrow}$$

Secondo passo

$$\frac{[x \mapsto 2, y \mapsto 3] \vdash d_3 \rightarrow^{(3)}}{[x \mapsto 2] \vdash d_2; d_3 \rightarrow [y \mapsto 3]; d_3 \rightarrow}$$

③ Elaborazione di d_3

$$\frac{[x \mapsto 2, y \mapsto 3] \vdash y + x \rightarrow^{(3)}}{[x \mapsto 2, y \mapsto 3] \vdash \text{const } z : \text{int} = y + x \rightarrow}$$

Valutazione di $y = x$

$$\left\{ \begin{array}{l} \frac{\overbrace{[x \mapsto 2, y \mapsto 3]}^p \vdash y \mapsto 3}{[x \mapsto 2, y \mapsto 3] \vdash y + x \mapsto 3 + x} \quad \text{perché } p(x) = 3 \\ \\ \frac{[x \mapsto 2, y \mapsto 3] \vdash x \mapsto 2}{[x \mapsto 2, y \mapsto 3] \vdash 3 + x \mapsto 3 + 2} \quad \text{perché } p(x) = 2 \\ \\ [x \mapsto 2, y \mapsto 3] \vdash 3 + 2 \mapsto 5 \quad \text{perché } 3 + 2 = 5 \\ \\ [x \mapsto 2, y \mapsto 3] \vdash y + x \mapsto^* 5 \quad \text{completato il passo ③} \end{array} \right.$$

$$\frac{[x \mapsto 2, y \mapsto 3] \vdash y + x \mapsto^* 5}{[x \mapsto 2, y \mapsto 3] \vdash d_2 \mapsto [z \mapsto 5]} \quad \text{completato il passo ④}$$

$$\frac{[x \mapsto 2, y \mapsto 3] \vdash d_3 \mapsto [z \mapsto 5]}{[x \mapsto 2] \vdash [z \mapsto 5]; d_3 \mapsto [y \mapsto 2], [z \mapsto 5]}$$

passo succ

$$[x \mapsto 2] \vdash [y \mapsto 3]; [z \mapsto 5] \mapsto [y \mapsto 3, z \mapsto 5]$$

||| questo completa l'elaborazione di d_2, d_3

$$[x \mapsto 2] \vdash d \mapsto^* [y \mapsto 3, z \mapsto 5] \quad \text{completo}$$

$$\frac{[x \mapsto 2] \vdash d \mapsto^* [y \mapsto 3, z \mapsto 5]}{\vdash [x \mapsto 2] \vdash d \mapsto [x \mapsto 2] \vdash [y \mapsto 3, z \mapsto 5]}$$

Ultimo passo

$$\vdash [x \mapsto 2] \vdash [y \mapsto 3, z \mapsto 5] \mapsto \underline{\underline{[y \mapsto 3, z \mapsto 5]}}$$

$$\Rightarrow \vdash d, d \mapsto^* [y \mapsto 3, z \mapsto 5]$$

$$= \text{Elab}(d_1; d) = [y \mapsto 3, z \mapsto 5]$$